

# AIMS 5740

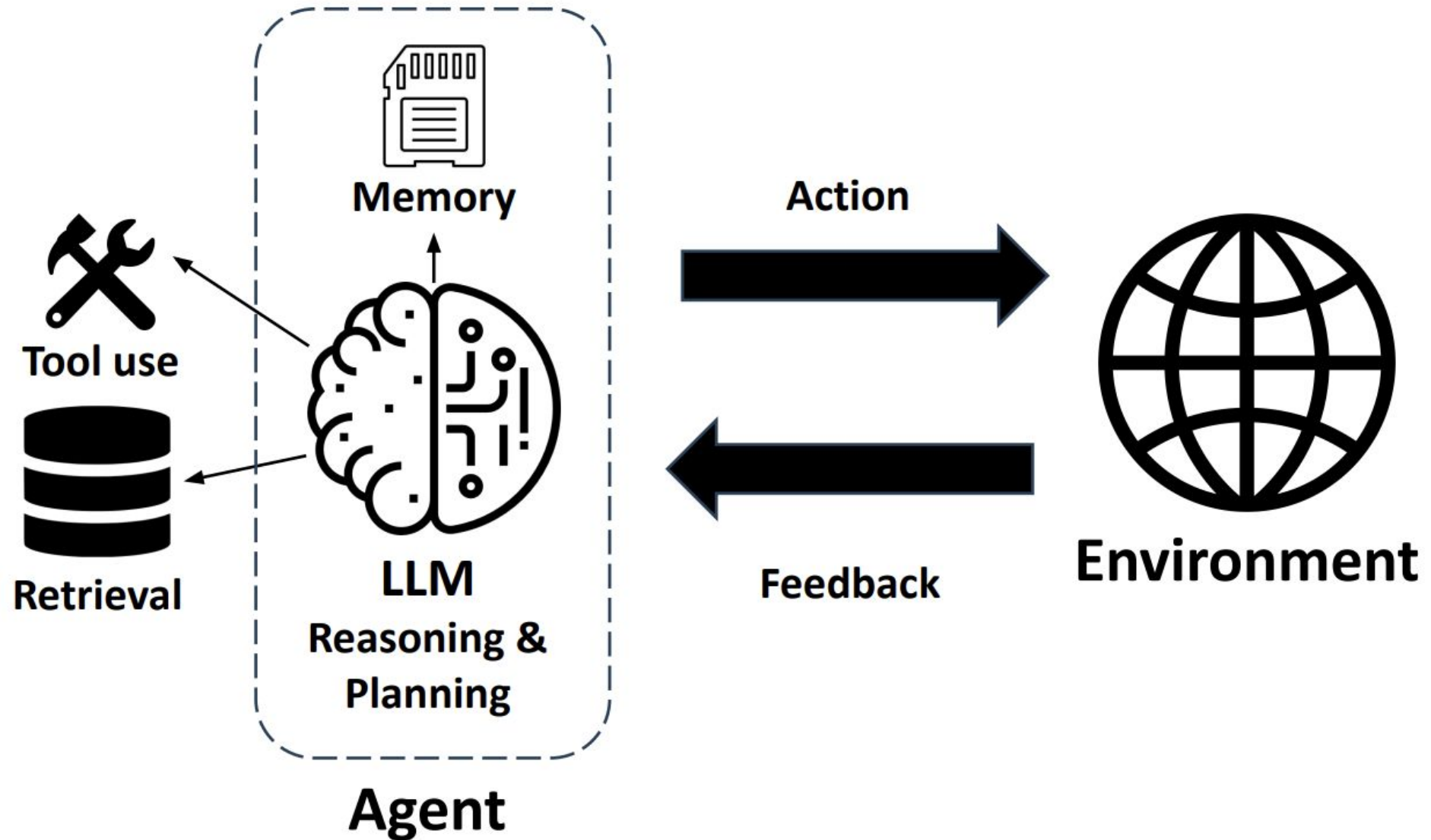
## Generative Artificial Intelligence

(2026 Term 2)

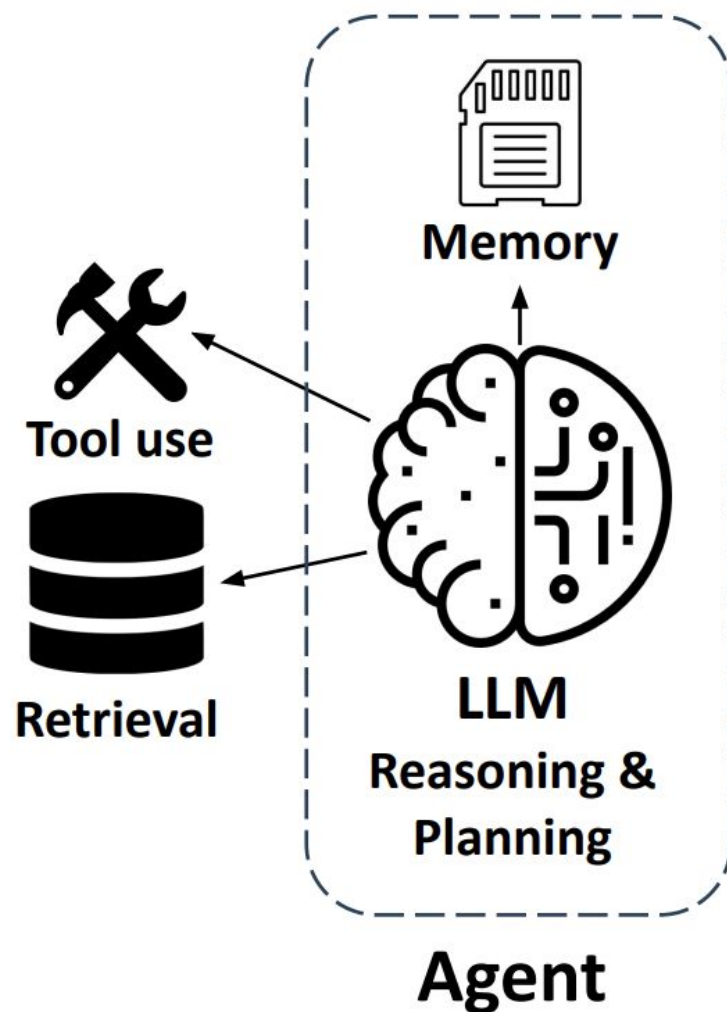


**Computer Science & Engineering**  
**The Chinese University of Hong Kong**

# LLM agents: enabling LLMs to interact with the environment

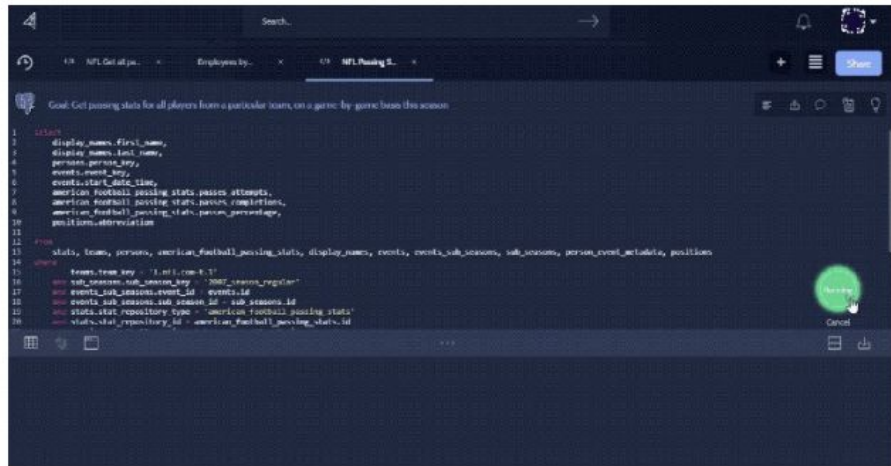


# Why empowering LLMs with the agent



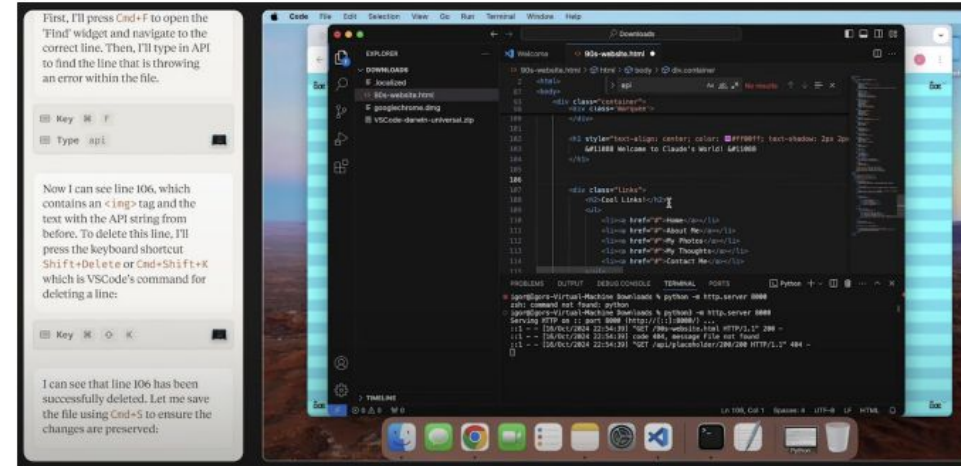
- Solving real-world tasks typically involves a trial-and-error process
- Leveraging external tools and retrieving from external knowledge expand LLM's capabilities
- Agent workflow facilitates complex tasks
  - Task decomposition
  - Allocation of subtasks to specialized modules
  - Division of labor for project collaboration
  - Multi-agent generation inspires better responses

# LLM agents transformed various applications



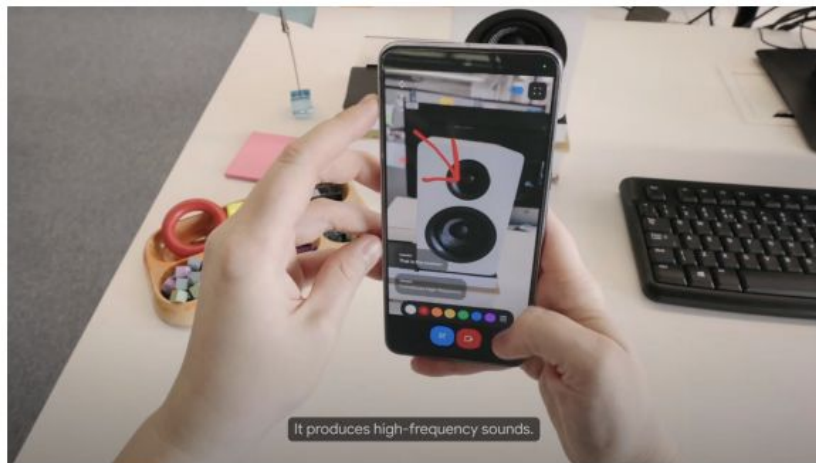
## Code generation

Cursor, GitHub Copilot, Devin, Google Jules...



## Computer use

Anthropic Claude, Google Jarvis, OpenAI Operator



## Personal assistant

Google Astra, OpenAI GPT-4o,...



## Robotics

Figure AI, Tesla Optimus, NVIDIA GR00T...

- Education
- Law
- Finance
- Healthcare
- Cybersecurity
- ...

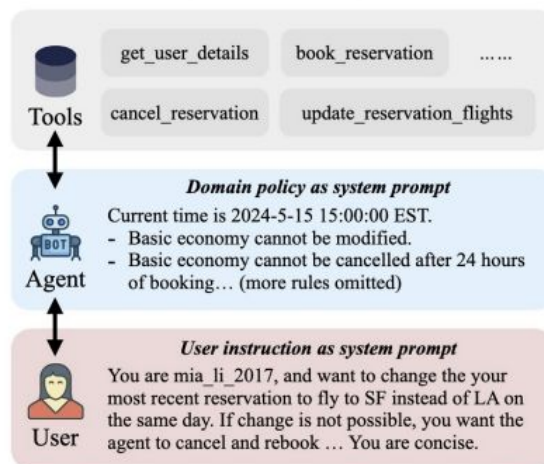


# Coding Agent

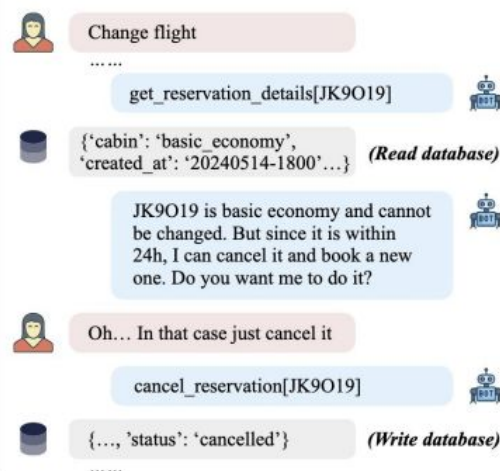
## API/functional calling for tool use

- Environment: software systems such as databases, app/web services...
- Observation space: API docs, system info, error messages and logs...
- Action space: function calls, error handling routines...

(a)  $\tau$ -bench setup



(b) Example trajectory in  $\tau$ -airline



```
{
  "order_id": "#W2890441",
  "user_id": "mei_davis_8935",
  "items": [
    {
      "name": "Water Bottle",
      "product_id": "8310926033",
      "item_id": "2366567022",
      "price": 54.04,
      "options": {
        "capacity": "1000ml",
        "material": "stainless steel",
        "color": "blue"
      }
    }
  ]
}
```

(a) An orders database entry in  $\tau$ -retail.

```
def return_delivered_order_items(
    order_id: str,
    item_ids: List[str],
    payment_method_id: str,
) -> str: ...

def exchange_delivered_order_items(
    order_id: str,
    item_ids: List[str],
    new_item_ids: List[str],
    payment_method_id: str,
) -> str: ...
```

(b) An API tool in  $\tau$ -retail.

```
## Return delivered order
- After user confirmation, the order status
  will be changed to 'return requested'...

## Exchange delivered order
- An order can only be exchanged if its
  status is 'delivered'...
```

(c) Domain policy excerpts in  $\tau$ -retail.

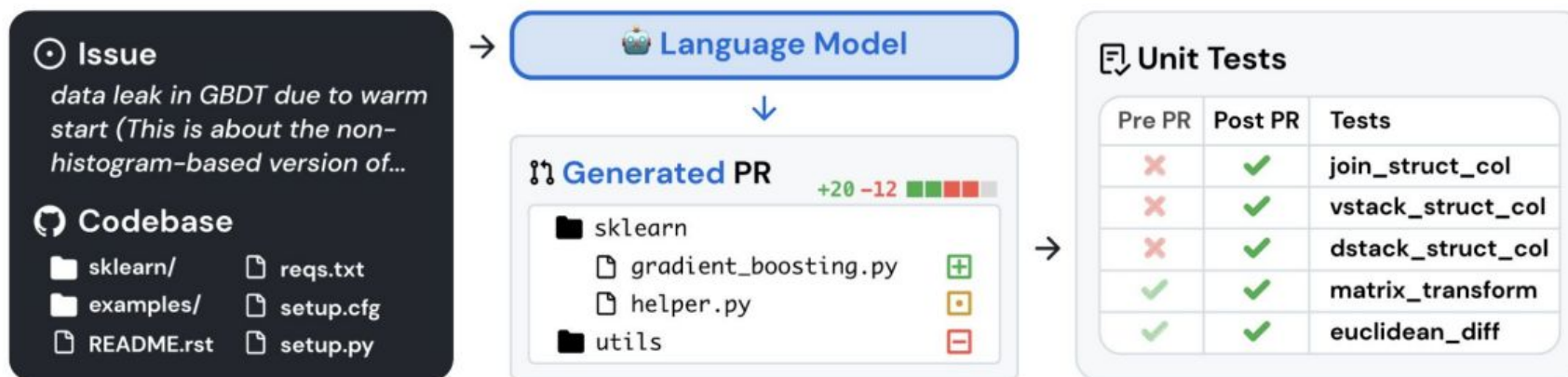
```
{
  "instruction": "You are Mei Davis in 80217. You want to return the water bottle, and exchange the pet bed and office chair to the cheapest version. Mention the two things together. If you can only do one of the two things, you prefer to do whatever saves you most money, but you want to know the money you can save in both ways. You are in debt and sad today, but very brief.",
  "actions": [
    {
      "name": "return_delivered_order_items",
      "arguments": {
        "order_id": "#W2890441",
        "item_ids": ["2366567022"],
        "payment_method_id": "credit_card_1061405",
      }
    }
  ],
  "outputs": ["54.04", "41.64"]
}
```

(d) User instruction ensures only one possible outcome.

# Coding Agent

## Project-level coding tasks

- Environment: project code repos, filesystems, IDEs...
- Observation space: code files, exe outputs, docs, errors, commit history...
- Action space: code edits, file search/view, test updates...





# Game Agent

## Digital games

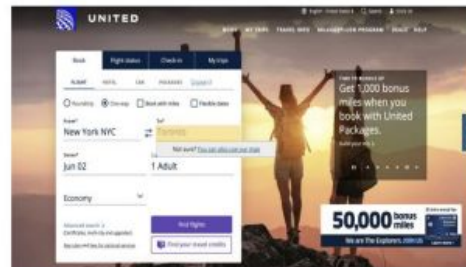
- Environment: game worlds/levels...
- Observation space: screenshots of game states, inventory, location...
- Action space: game controls (e.g., drop, move, attack, resource management...)



# Web/app agents

## Web/app use

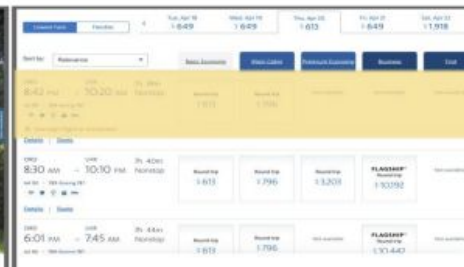
- Environment: web browsers/apps
- Observation space: screenshots, DOM trees, HTML, historical actions...
- Action space: browser/app controls (e.g., click, type, scroll, drag, hover...)



(a) Find one-way flights from New York to Toronto.



(b) Book a roundtrip on July 1 from Mumbai to London and vice versa on July 5 for two adults.



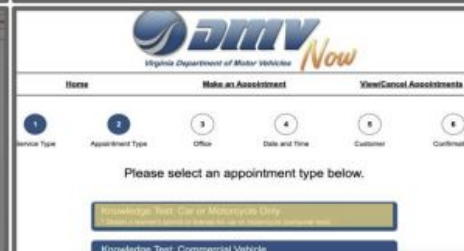
(c) Find a flight from Chicago to London on 20 April and return on 23 April.



(d) Find Elon Musk's profile and follow, start notifications and like the latest tweet.



(e) Browse comedy films streaming on Netflix that was released from 1992 to 2007.



(f) Open page to schedule an appointment for car knowledge test.

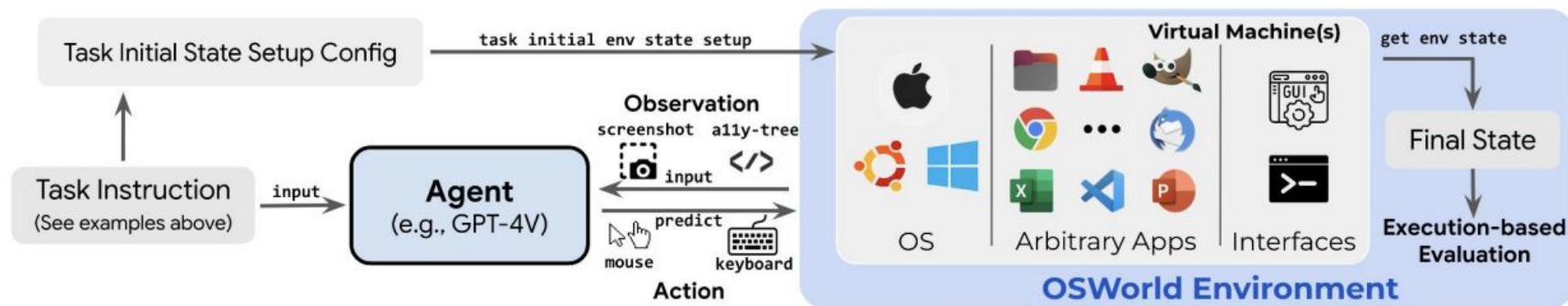


# Computer use agents

Computer use for universal digital tasks

- Environment: desktop operating systems
- Observation space: desktop screenshots, a11y trees, historical actions...
- Action space: keyboard/mouse controls (e.g., click, type, drag, shortcuts)

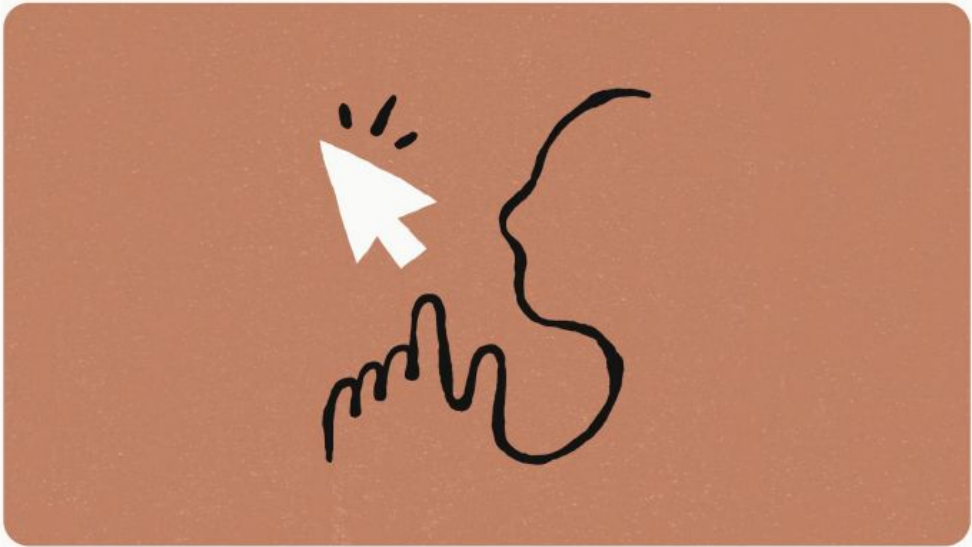
Task instruction I: Update the bookkeeping sheet with my recent transactions over the past few days in the provided folder.



# Computer use agents

## Introducing computer use, a new Claude 3.5 Sonnet, and Claude 3.5 Haiku

Oct 22, 2024 • 5 min read



| Category     | Claude 3.5 Sonnet (New) - 15 steps |               | Claude 3.5 Sonnet (New) - 50 steps |               | Human Success Rate [3] |
|--------------|------------------------------------|---------------|------------------------------------|---------------|------------------------|
|              | Success Rate                       | 95% CI        | Success Rate                       | 95% CI        |                        |
| OS           | 54.2%                              | [34.3, 74.1]% | 41.7%                              | [22.0, 61.4]% | <b>75.00 %</b>         |
| Office       | 7.7%                               | [2.9, 12.5]%  | 17.9%                              | [11.0, 24.8]% | <b>71.79 %</b>         |
| Daily        | 16.7%                              | [8.4, 25.0]%  | 24.4%                              | [14.9, 33.9]% | <b>70.51 %</b>         |
| Professional | 24.5%                              | [12.5, 36.5]% | 40.8%                              | [27.0, 54.6]% | <b>73.47 %</b>         |
| Workflow     | 7.9%                               | [2.6, 13.2]%  | 10.9%                              | [4.9, 17.0]%  | <b>73.27 %</b>         |
| Overall      | 14.9%                              | [11.3, 18.5]% | 22%                                | [17.8, 26.2]% | <b>72.36 %</b>         |

Anthropic recent computer use agent results of OSWorld

# Robotic agents

## Robotics for physical interaction

- Environment: physical world spaces
- Observation space: visual input, sensor readings, physical states, proprioception...
- Action space: motor controls (e.g., move, grasp, manipulate...)





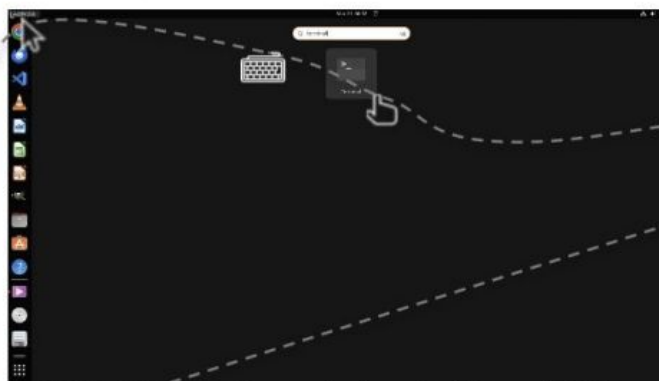
# Agent Data

## Task goal aligned trajectories (observation-action pairs)

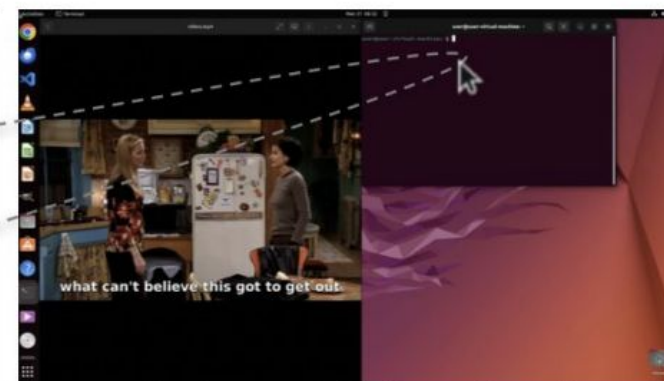
Task Instruction: I downloaded an episode of Friends to practice listening, but I don't know how to remove the subtitles. Please help me remove the subtitles from the video and export it as "subtitles.srt" and store it in the same directory as the video.



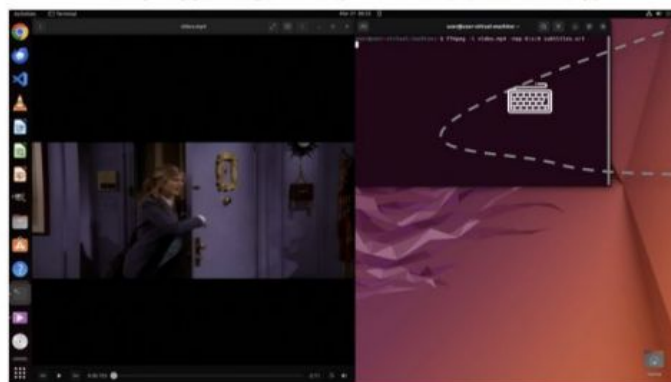
Step 1: `pyautogui.click(activities_x, activities_y)`



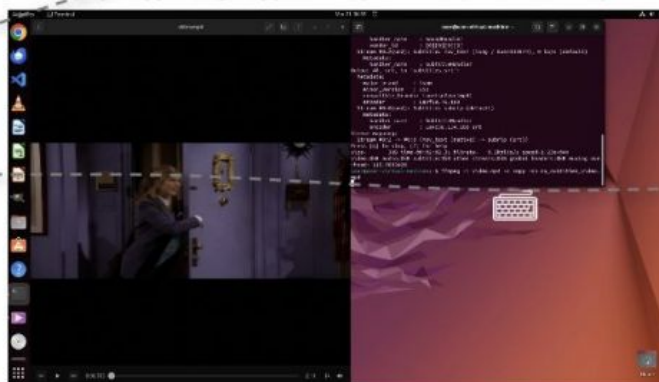
Step 2: `pyautogui.typewrite('terminal', interval=0.5)`



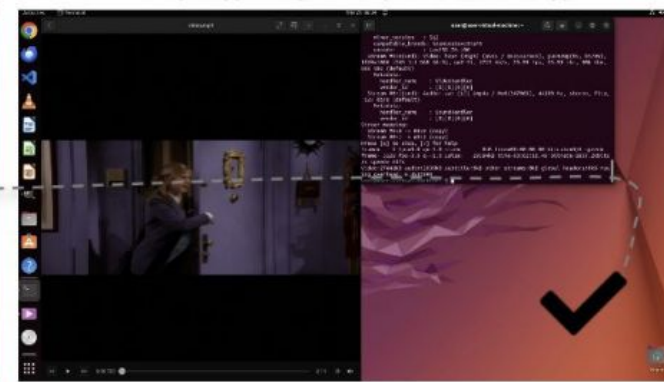
Step 3: `pyautogui.click(focus_x, focus_y)`



Step 4: `pyautogui.typewrite('ffmpeg -i video.mp4 -map 0:s:0 subtitles.srt', interval=0.5)`



Step 5: `pyautogui.typewrite('ffmpeg -i video.mp4 -c copy -sn no_subtitles_video.mp4', interval=0.5)`

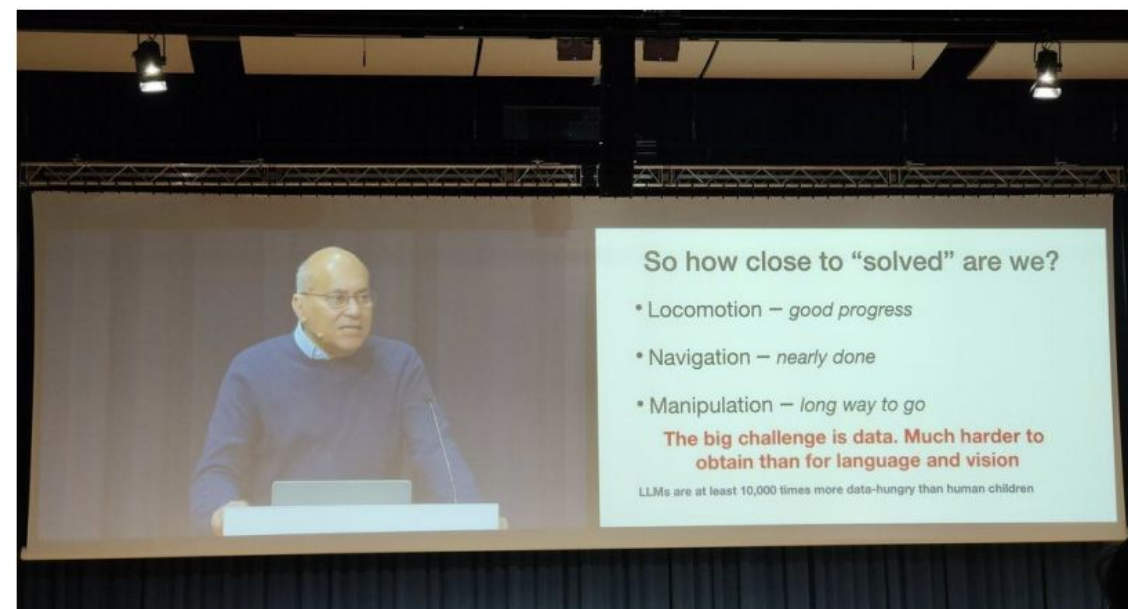


Step 6: Done

# Agent Data

Agent data is hard to get directly from internet-scale text and videos due to **embodiment**.

- Complex data collection infrastructure
- Complex observation-action interaction in diverse environments
- Goal aligned trajectories



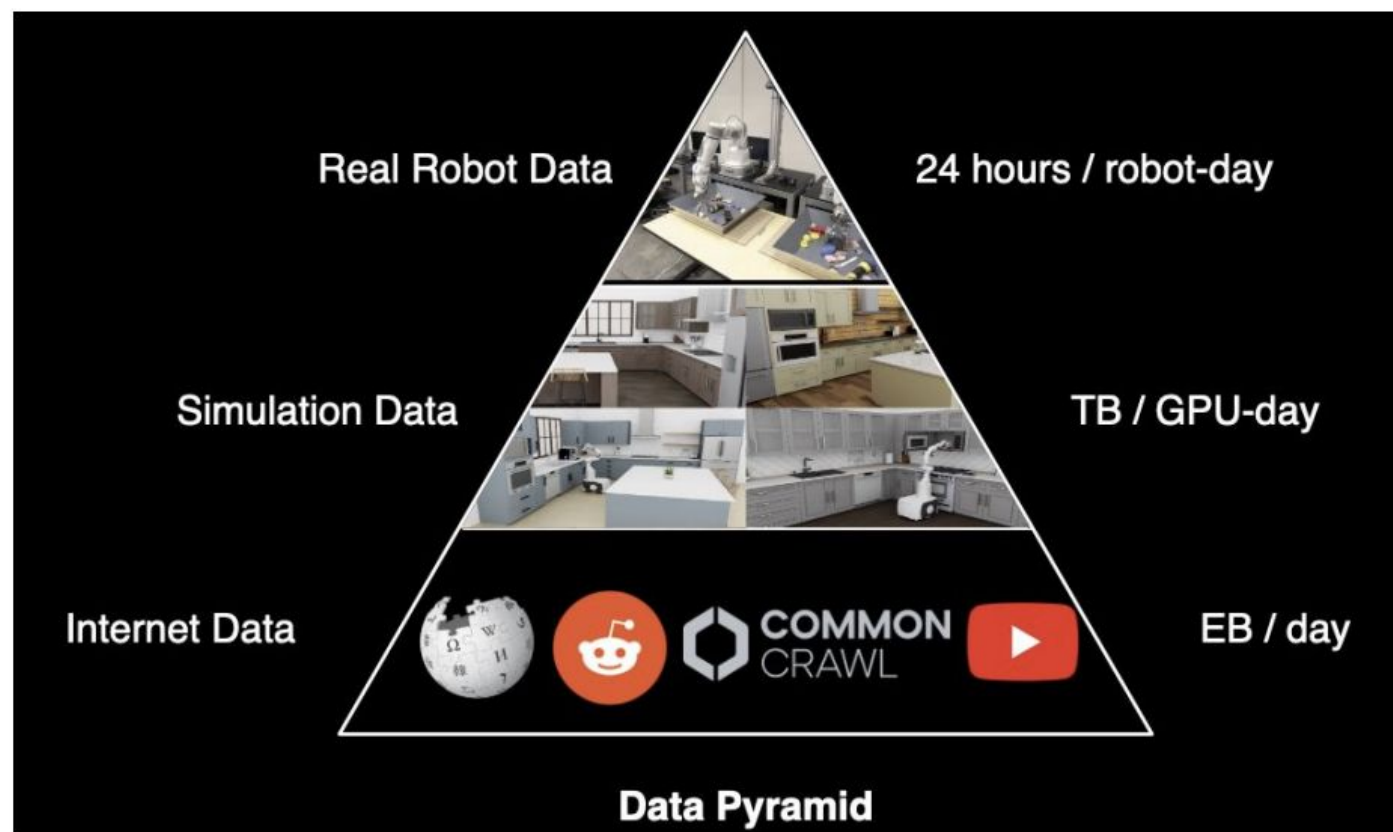
Jitendra’s talk @ CoRL 2024



# Scaling agent data

Scaling agent data through

- Human demonstrations
- Synthesis/simulation
- Internet-scale data





# Agent evaluation: in the era of LLMs is hard



**Andrej Karpathy** ✓

@karpathy



Nice, a serious contender to [@lmsysorg](#) in evaluating LLMs has entered the chat.

LLM evals are improving, but not so long ago their state was very bleak, with qualitative experience very often disagreeing with quantitative rankings.

This is because good evals are very difficult to build - at Tesla I probably spent 1/3 of my time on data, 1/3 on evals, and 1/3 on everything else. They have to be comprehensive, representative, of high quality, and measure gradient signal (i.e. not too easy, not too hard), and there are a lot of details to think through and get right before your qualitative and quantitative assessments line up. My goto pointer for some of the fun subtleties is probably the Open LLM Leaderboard MMLU writeup: [github.com/huggingface/bl...](https://github.com/huggingface/bl...)

# | Agent evaluation is even more challenging

## Challenges in agent evaluation

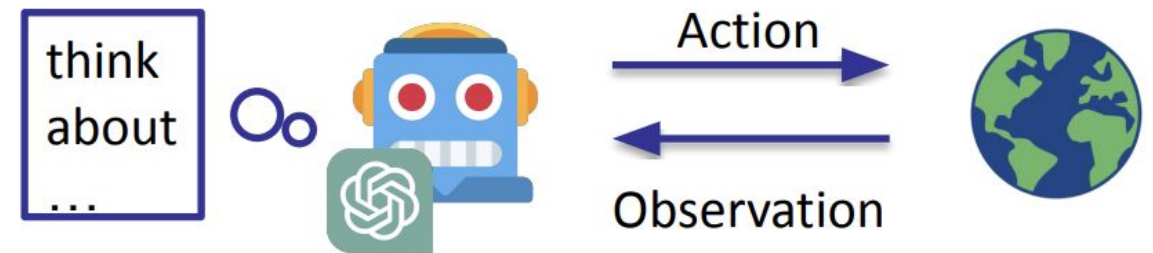
- Real-world environmental setup complexity
- Task coverage
- Open-ended success criteria
  - Multiple valid solution paths
  - Cannot script evaluation metrics, need for human judgment
- Evaluation beyond task success

## Agent evaluation

- via benchmarks
- via LLMs/VLMs
- via crowdsourcing

# Reasoning for Agent

- For humans: various mental processes
- For LMs: intermediate generation
- For agents: internal actions
  - Why is it helpful?
  - Why is it special?
  - Why is it now?





# Reasoning is helpful for Agent

Obs  $t$

You are cooking a dish and seeing salt is out...

**Reasoning**

“The dish should be savory, and since salt is out, I should find the soy sauce instead. It is in the cabinet to my right...”

Act  $t$

Turn right

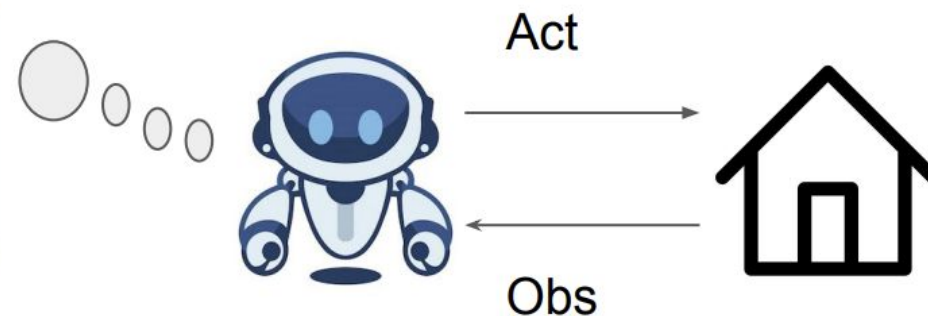
Obs  $t+1$

You see a cabinet and a table...

Act  $t+1$

Open cabinet

.....



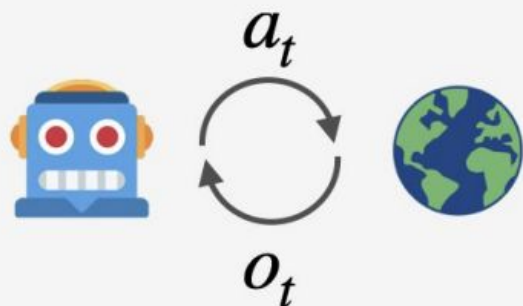
## Reason to act

1. Generalization
2. Alignment

Observation to Action mapping could be hard

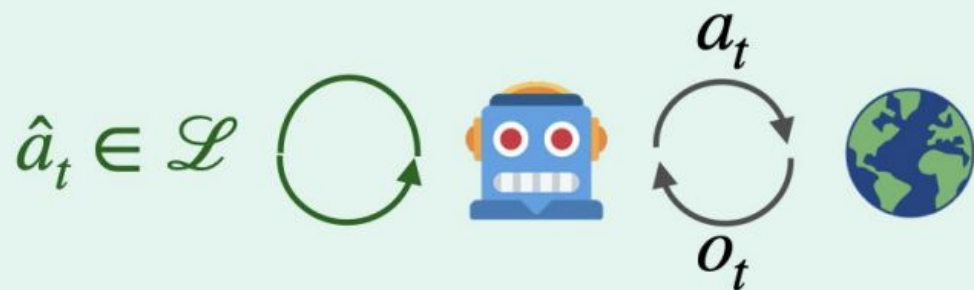
# Why is reasoning special for agents?

**Traditional agents:** action space  $A$  defined by the environment



- **External feedback**  $o_t$
- Agent context  $c_t = (o_1, a_1, o_2, a_2, \dots, o_t)$
- Agent action  $a_t \sim \pi(a | c_t) \in A$

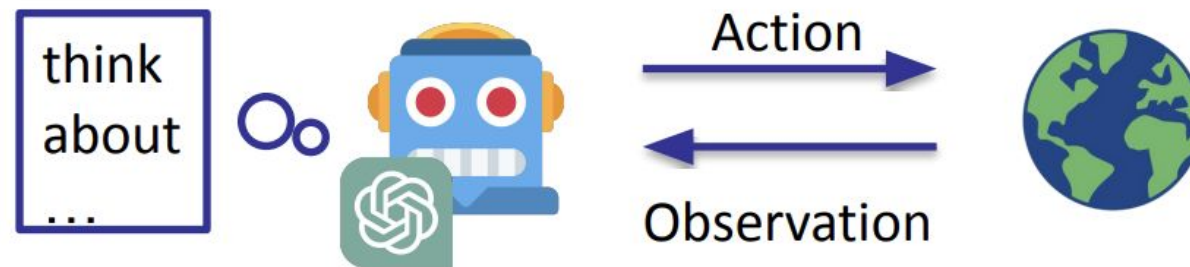
**ReAct:** action space  $\hat{A} = A \cup \mathcal{L}$  augmented by reasoning



- $\hat{a}_t \in \mathcal{L}$  can be any language sequence
- Agent context  $c_{t+1} = (c_t, \hat{a}_t, a_t, o_{t+1})$
- $\hat{a}_t \in \mathcal{L}$  only updates **internal context**

# Reasoning: Takeaways

- Reasoning as internal actions for agents
  - No external feedback, just change internal context
  - Infinite space -> requires strong priors to navigate
- Reasoning guides acting, acting updates reasoning
- Action planning approaches (e.g. tree search) can improve reasoning

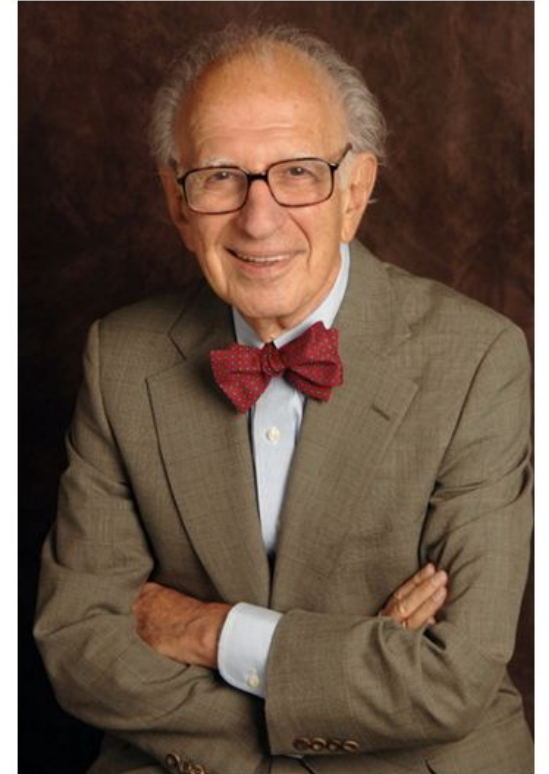




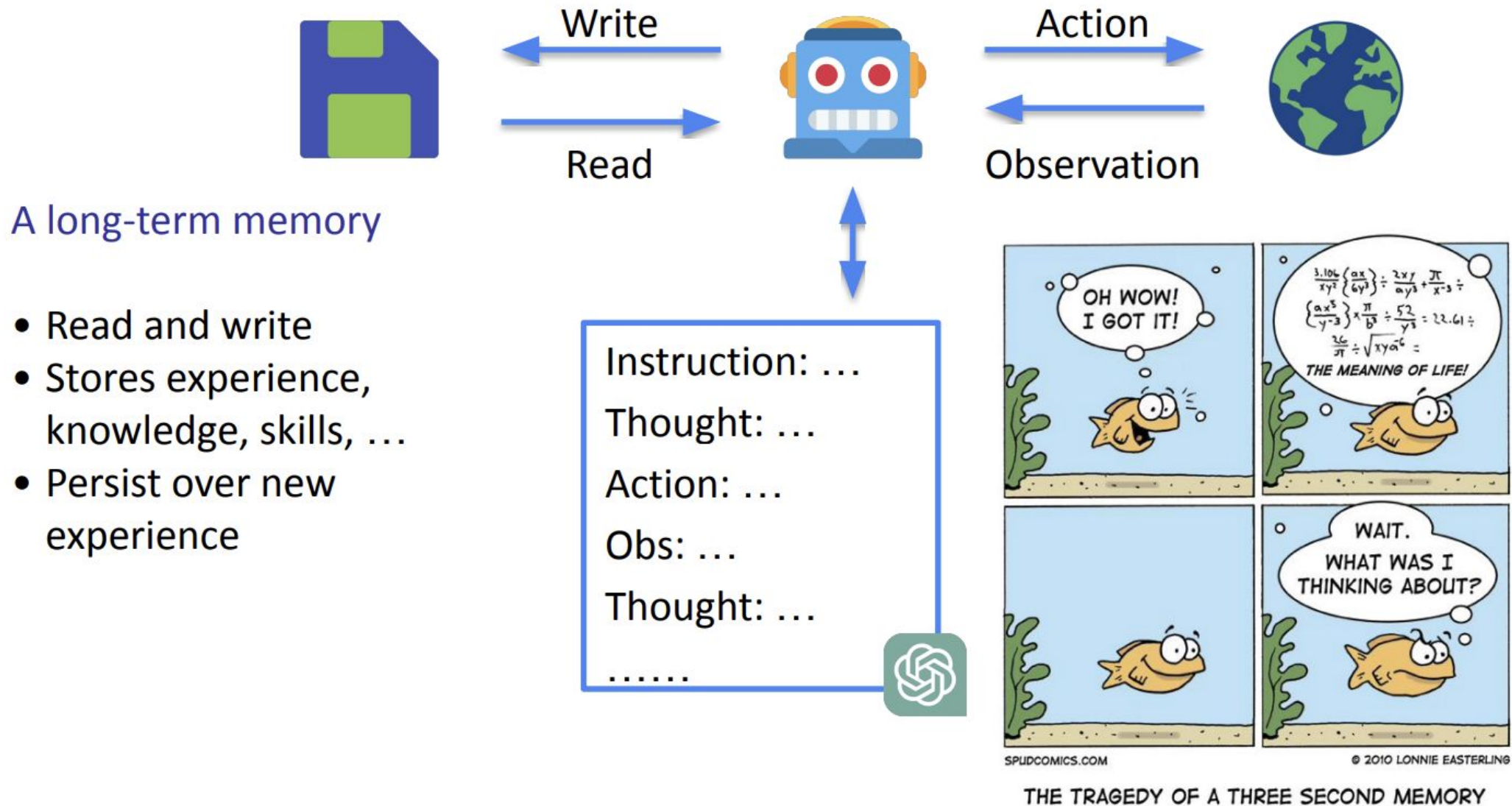
# | Agent Memory

Memory is everything. Without it  
we are nothing.

— Eric Kandel

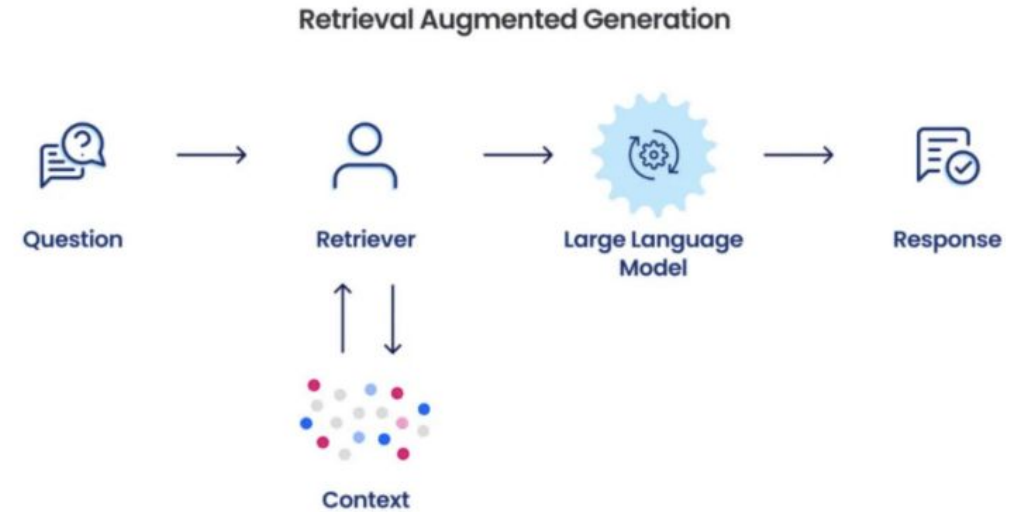


# Agent Memory



# Retrieval and RAG

- We can think of the retrieval corpus as “read-only” LTM
  - Written by others (e.g., Wikipedia editors), not the agent itself
  - Retrieve[query]: a read action
- Limitations
  - Can only live “others’ experience”, which might not be optimal for the agent
  - The way corpus is written might not be optimal for agent usage
- Agent memory: also be able to autonomously write to it!





# Long-term Memory: Content

| Type by content   | Definition        | Examples                                 |
|-------------------|-------------------|------------------------------------------|
| Episodic memory   | Stores experience | Generative agents<br>[Park et al., 2023] |
| Semantic memory   | Stores knowledge  |                                          |
| Procedural memory | Stores skills     | Voyager [Wang et al., 2023]              |

- Note: here we categorize based on memory content, which is
  - Inspired by [human long-term memory systems](#)
  - orthogonal to implementations

# Memory: Takeaways

- Language agents interact with external environments and internal memories (information-storing devices)
  - Interact with short-term memory (context window): reasoning
  - Interact with long-term memory (LLM weights, event logs, codebase, prompt library, etc.): retrieving and **learning**
- Exercise question: what's the difference between external environment vs internal memory then? (Hint: check CoALA)



# Multi-Agent: Motivation

- Single agent is not strong enough!
- An easy-to-use prompting for effectiveness
- Test of time compute
  - Parallel & easy to scale
- Different agents represent different expertise
- Decentralized control & Privacy-persevering
  - Maintain data access control



# ■ Key Principles of Multi-Agent

## ❑ Agent initialization

- ❑ Profiling via persona or expertise; task decomposition

## ❑ Orchestration process

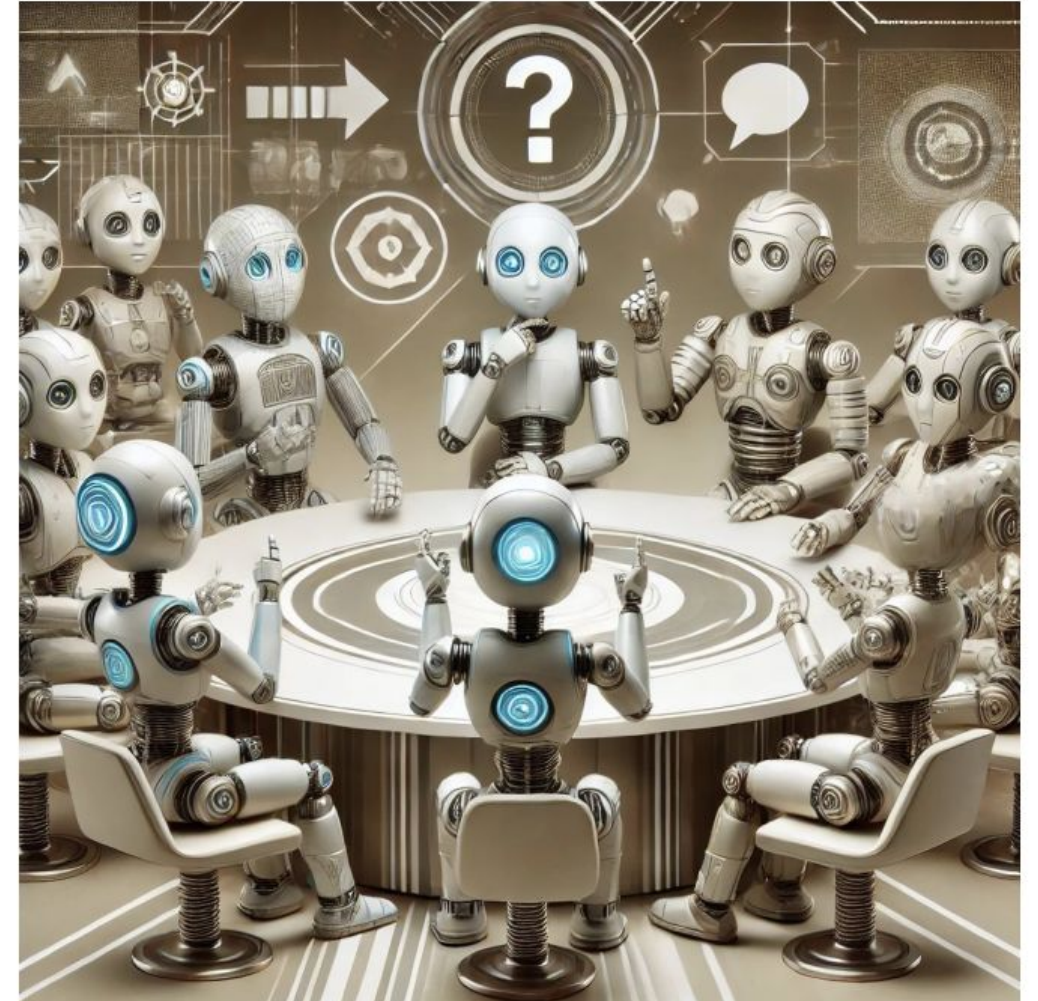
- ❑ Coordinate different agents

## ❑ Agent team optimization

- ❑ Optimize an existing agent team; agent selection

# Coordinate Multi-Agents via LLM Debate

- Multiple LLM agent instances
- Propose and debate their individual responses
- Reason over multiple rounds for final answer



# Coordinate Multi-Agents via LLM Debate

**Basic idea:** prompt multiple LM instances to debate each other to find the answer.

**Question:** Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

## Round 1

**ChatGPT:** Before the restart, Carla was able to download 40% of the file, which is  $0.4 * 200 \text{ GB} = 80 \text{ GB}$ . At a rate of 2 GB/minute, it would take her:  $\text{time} = 80 \text{ GB} \div 2 \text{ GB/minute} = 40 \text{ minutes}$ .

Because of the restart, Carla had to start over. This means she had to download the entire 200 GB file. The time it takes to download the remaining 120 GB is:  $\text{time} = 120 \text{ GB} \div 2 \text{ GB/minute} = 60 \text{ minutes}$ .

But Carla also had to wait 20 minutes for the Windows update to install, so the total time it took to download the file is:  $\text{total time} = 40 \text{ minutes} + 60 \text{ minutes} + 20 \text{ minutes} = 120 \text{ minutes}$ . ❌

**Bard:** Carla is downloading a 200 GB file at a speed of 2 GB/minute. This means that it will take her  $200 / 2 = 100 \text{ minutes}$  to download the file.

However, Windows forces a restart to install updates, which takes 20 minutes. This means that Carla will have to restart the download from the beginning. Since Carla has already downloaded 40% of the file, she only needs to download  $200 - 40 = 160 \text{ GB}$ . At a speed of 2 GB/minute, it will take Carla  $160 / 2 = 80 \text{ minutes}$  to download the file.

Adding the time for the Windows restart, it will take Carla  $80 + 20 = 100 \text{ minutes}$  to download the file. ❌

## Round 2

**ChatGPT :** Based on the calculations provided by the other agent, the correct answer is: Before the restart, Carla was able to download 40% of the file, which is  $0.4 * 200 \text{ GB} = 80 \text{ GB}$ . At a rate of 2 GB/minute, it would take her:  $\text{time} = 80 \text{ GB} \div 2 \text{ GB/minute} = 40 \text{ minutes}$

However, because of the restart, Carla had to start over. This means she had to download the entire 200 GB file. The time it takes to download the entire 200 GB file is:  $\text{time} = 200 \text{ GB} \div 2 \text{ GB/minute} = 100 \text{ minutes}$

And since Carla had to wait 20 minutes for the Windows update to install, the total time it took to download the file is:  $\text{total time} = 40 \text{ minutes} + 100 \text{ minutes} + 20 \text{ minutes} = 160 \text{ minutes}$ . ✅



# Coordinate Multi-Agents via LLM Debate

|                              | # responses | GSM8K       |
|------------------------------|-------------|-------------|
| Standard Prompting           | 1           | 76.7        |
| Self-Consistency             | 3           | 82.5        |
| Multi-Agent Debate (round 1) | 6           | 83.2        |
| Self-Consistency             | 6           | 85.3        |
| Multi-Agent Debate (round 2) | 9           | 83.0        |
| Self-Consistency             | 9           | <b>88.2</b> |

❖ **Multi-agent debate**: prompt the LLM to review multiple responses and give an updated one.

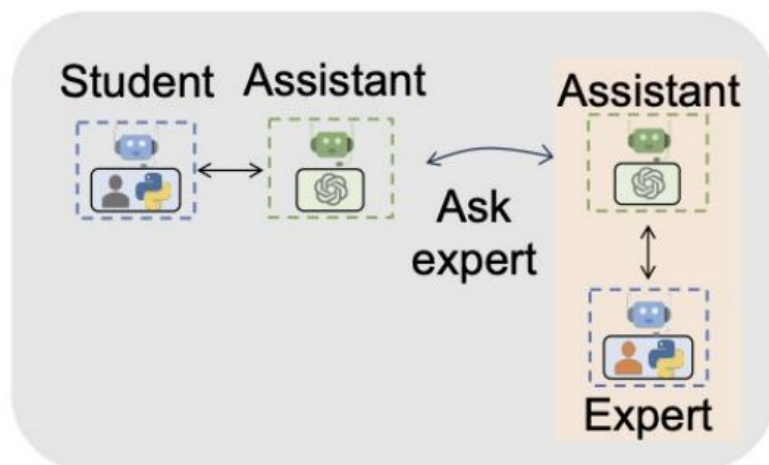
Vs.

❖ **Self-consistency** selects the response with the most common final answer.

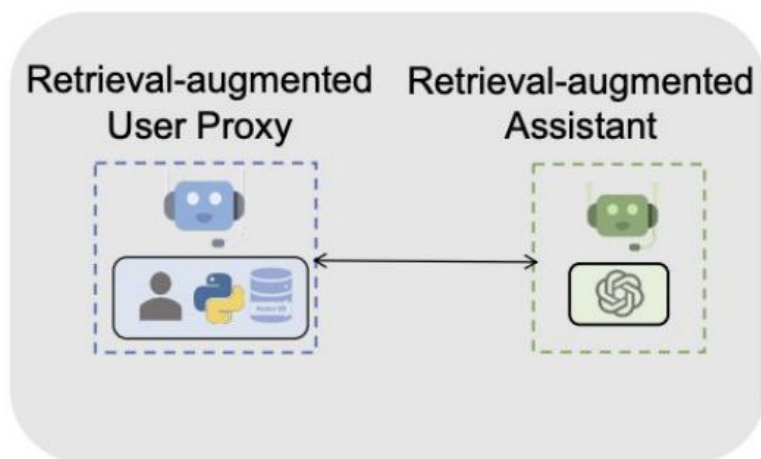
❖ With the right prompt, Multi-agent does not outperform SC

❖ Without a good evaluator, multi-agent debate does not effectively utilize the token budget.

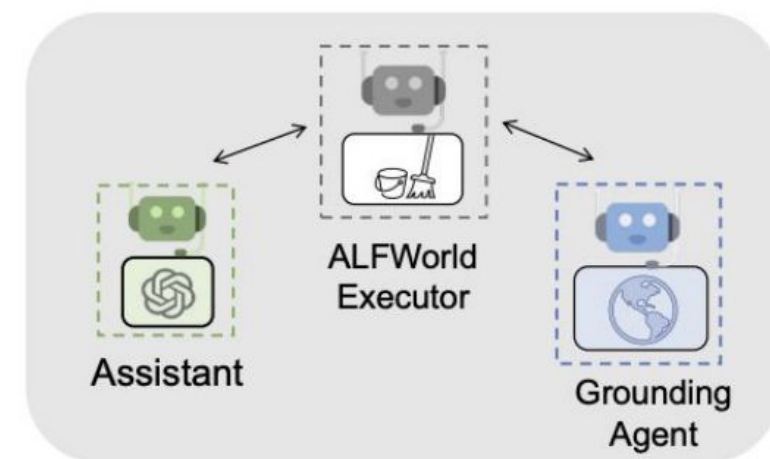
# Coordinate with Multi-Agent Conversation



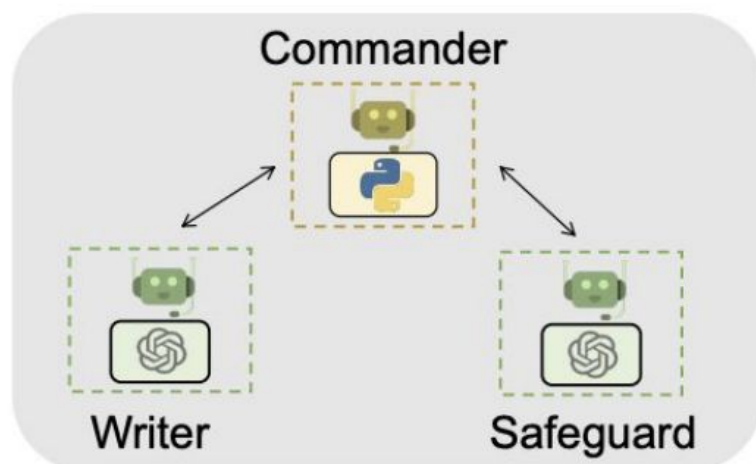
A1. Math Problem Solving



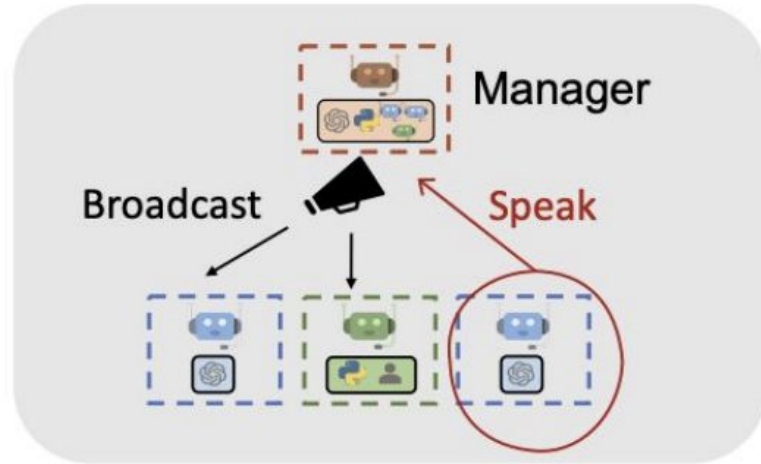
A2. Retrieval-augmented Q&A



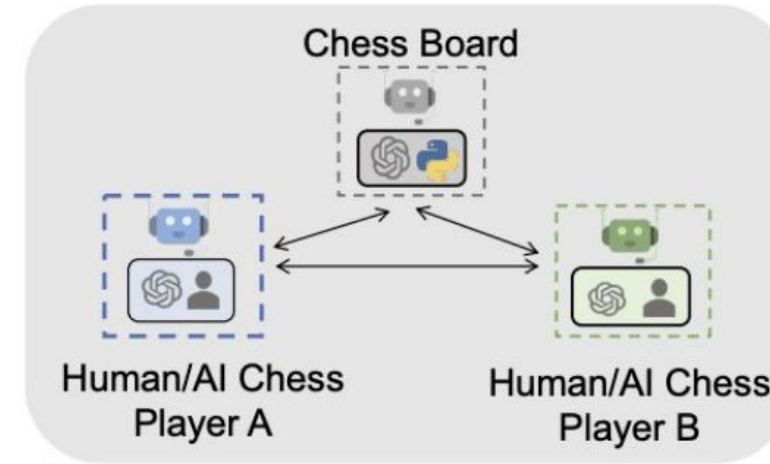
A3. Decision Making in Embodied Agents



A4. Supply-Chain Optimization



A5. Dynamic Task Solving with Group Chat



A6. Conversational Chess

# Thanks

## Q&A

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